Computational modeling of grain growth for additively manufactured components using Monte Carlo simulation

Wongel Nadew

November 2025

Abstract

Additive manufacturing (AM) enables the fabrication of complex geometries that are difficult or impossible to achieve through traditional manufacturing methods. It produces a complex microstructure depending on the processing conditions. However, characterizing the process-microstructure relation of additively manufactured components remains elusive. To bridge this knowledge gap, computational modeling tools offer a powerful way to simulate the process environment and grain growth, and to visualize them effectively. This project utilizes SPPARKS, an open-source Monte Carlo simulator, to conduct lattice-based simulations of grain structure formation. Specifically, a mesoscale Potts-kinetic Monte Carlo method is employed to investigate the relationship between melt pool geometry and temperature profiles with the grain size and shape in metal AM structures using the Direct Deposition Method. To demonstrate the model's effectiveness, simulations of 304L stainless steel produced via Direct Deposition are validated against experimental EBSD data. Further investigations of the influence of processing parameters on grain structure growth are also conducted.

1 Introduction

1.1 Overview of metal additive manufacturing methods

Metal additive manufacturing offers numerous advantages over traditional manufacturing processes, particularly by granting design freedom to create complex parts with sophisticated geometry [10]. With the development of additive manufacturing technology, a new manufacturing paradigm has emerged, enabling control over the microstructure of components [2]. Based on how powder is supplied to build a component, metal additive manufacturing can be categorized into two main classes. These categories are Powder Bed Fusion (PBF) and Laser Engineered Net Shaping (LENS). LENS methods use a carrier gas stream that transports powder and delivers it via a nozzle to the melt pool, which is simultaneously being melted by a laser [4]. In the LENS method, both the powder delivery nozzle and the coaxial laser assembly move in tandem.

1.2 Computational modeling of metal additive manufacturing

Computational modeling of metal additive processes continues to be a challenge because the AM process is an iterative process in which optimal parameters are identified through a series of trials [11]. The inherent complexity of the process involved in AM, including layer-by-layer material deposition, fusion, and solidification, has posed a challenge for finite element simulations. Furthermore, varying process parameters yield diverse microstructures, and this diversity can be attributed to non-uniform local solidification behavior occurring during the printing process [10]. In addition to the solidification process, different scanning strategies produce significantly different grain structures [1]. Consequently, there is a need for a computational modeling tool capable of replicating the layer-by-layer material processing, and Monte Carlo simulations are ideal methods for estimating the influence of these input parameters on the final microstructures of the printed material.

One of the major assumptions underlying the Monte Carlo simulation used here is that the melt pool shape is assumed to be constant since the thermal and microstructural evolution models are decoupled. To conduct a grain growth study for AM parts, we need to perform finite element simulations and determine the melt pool geometry. We relay the geometric information to a Monte Carlo model and simulate the grain growth. This work follows the steps described by Trageser et. al.[11] to simulate microstructure growth of 304L stainless steel manufactured using the Direct Deposition Method. Simulated microstructures are compared to EBSD grain maps. Further investigations on how scanning speed and melt pool size influence grain growth are also conducted.

2 Methods

2.1 Monte Carlo (MC) Potts Simulations

In statistical mechanics, Monte Carlo (MC) simulations are a stochastic algorithm that uses random numbers to evolve a system from state to state [5]. The Potts model is implemented in the SPPARKS [7, 6] simulation package developed by Sandia National Laboratory. The Monte Carlo Potts Model is a lattice-based application to simulate curvature-driven grain growth [3]. It is used to model the grain growth of polycrystalline microstructure by assigning integer lattice "spins" to each lattice site. Neighboring sites of similar spin integer make up a grain. These lattice sites represent a discrete bit of material. The Potts model operates by minimizing the total grain boundary energy (E), which is given by:

$$E = \sum_{i}^{N} \sum_{j}^{n} (1 - \delta(q_i, q_j)) \tag{1}$$

where N is the total number of lattices sites, and n is the number of lattice sites ($n = 26$ is used throughout this report). q_i and q_j are the spins of each site and the neighboring sites, and δ is the Kronecker delta function, which assumes a value of 1 when q_i and q_j have the same spin or 0 when they differ. Neighbors with dissimilar spins contribute to the total energy of the system.

The grain boundary energy is minimized through curvature-driven grain growth by evolving the spins of lattice sites using random numbers. Spin reassignment at these sites is either accepted or rejected using the Metropolis algorithm by comparing a random number generated from a uniform distribution over the interval $[0, 1]$ to the acceptance probability, P. The acceptance probability for a spin reassignment MC move is calculated by:

$$P = \begin{cases} exp\left(\frac{-\Delta E}{k_B T_s}\right), & if \quad \Delta E > 0 \\ 1, & if \quad \Delta E \leq 0 \end{cases} \tag{2}$$

where ΔE is the change of grain boundary energy, k_B is Boltzmann's constant and T_s is the simulation background temperature. If the spin reassignment reduces the energy ($\Delta E \leq 0$), the MC reassignment is automatically accepted. If the change in energy is positive ($\Delta E > 0$), then the move or spin reassignment is only accepted if the random number drawn is less than P (the probability acceptance rate). The simulation advances in unit increments of the Monte Carlo step (MCS), which corresponds to the evaluation of spin update across all sites within the simulation domain.

2.2 MC Potts model for Metal Additive Manufacturing

The MC Potts method has been successfully applied to welding simulations modeling the microstructural evolution during a welding pass using SPPARKS software [9]. Given the process similarities between welding and additive manufacturing, the same simulation principles have also been extended to AM applications [8]. This method simulates the rastering of the melt pool and its accompanying heat-affected zone (HAZ). Rastering is implemented in a layer-by-layer fashion, which is combined to form the overall geometry of the component being printed. The initial implementation models the melt pool as a double ellipsoid. An improvement in representing the melt pool was later developed by using a Bezier curve to trace out the melt pool surface [11]. A Bezier curve is a parametric curve constructed using Bernstein polynomials as basis functions. A Bernstein polynomial of degree n and parametrized with a variable u is defined as:

$$\mathbf{B}_{i,n} = \frac{n!}{i!(n-i)!} u^i (1-u)^{n-i}, \quad i \in [0, 1, ..., n], \quad u \in [0, 1]. \tag{3}$$

To construct the melt pool, two Bėzier curves are employed and parametrized by u. For the top surface, control points $\mathbf{P}_i$ are used to fit the top Bezier curve, $\mathbf{C}^t(u)$. For the spline curve, control points $\mathbf{Q}_i$ are used so that $\mathbf{C}^{sp}(u)$ approximates the longitudinal boundary of the melt pool. The two fourth-order Bezier curves (with $n = 4$) are shown in Figure 1 along with the control points $\mathbf{P}_i$ and $\mathbf{Q}_i$, and calculated as:

$$\begin{aligned} \mathbf{C}^t(u) &= \sum_{i=0}^{N} \mathbf{P}_i B_{i,n} \\ \mathbf{C}^{sp}(u) &= \sum_{i=0}^{N} \mathbf{Q}_i B_{i,n} \end{aligned} \tag{4}$$

The simulation parameters selected are calibrated against experimental measurements. The control points ($\mathbf{P}_i$ and $\mathbf{Q}_i$) given in Table 1 are selected to fit the melt

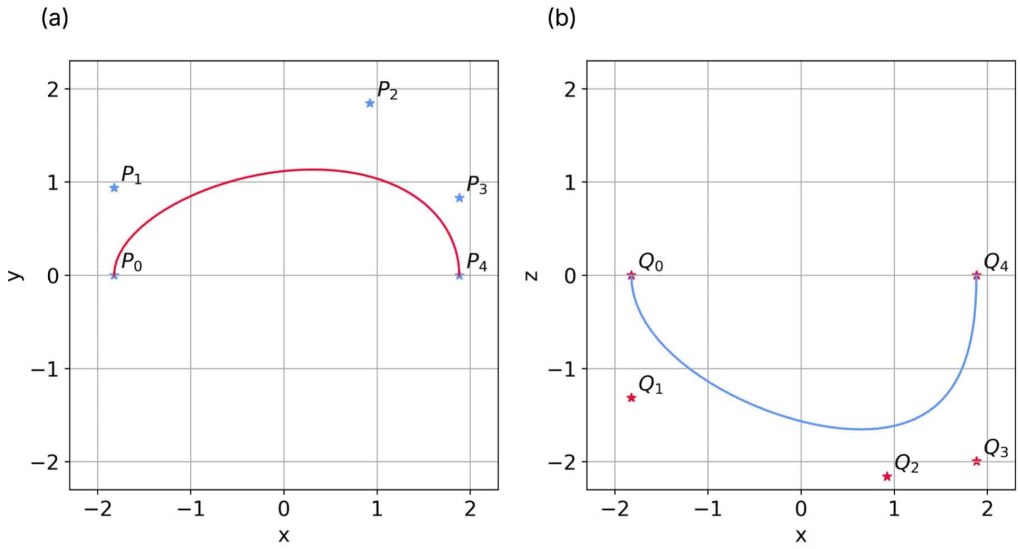


Figure 1: (a) The top surface Bezier curve fit using $\mathbf{P}_i$ as control points. (b) The subsurface Bezier curve to fit the spline curve using $\mathbf{Q}_i$ as control points.

Table 1: Fitted Bezier curve control points in Figure 1 for the simulated melt pool geometry depicted in Fig 2. Adapted from [11]

(a) Top surface curve control points for melt pool fit.

Steel	x(mm)	y(mm)
P_0	-1.824	0.000
P_1	-1.824	0.938
P_2	0.923	1.845
P_3	1.882	0.830
P_4	1.882	0.000

(b) Spine curve control points for melt pool fit.

Steel	x(mm)	z(mm)
Q_0	-1.824	0.000
Q_1	-1.824	-1.313
Q_2	0.923	-2.157
Q_3	1.882	-1.993
Q_4	1.882	0.000

pool geometry well. Another modification to the standard Potts model is that the microstructural evolution is limited to the immediate vicinity of the melt pool. Grain structures are not expected to grow far from the melt pool due to the negligible grain mobility for locations with low temperature. To incorporate this effect, grain boundary mobility ($M(T)$), which is a function of local temperature (T), is introduced. The acceptance probability stated in Equation 2 is updated to the following form:

$$P = \begin{cases} M(T)\, exp\left(\frac{-\Delta E}{k_B T_s}\right), & if \quad \Delta E > 0 \\ M(T), & if \quad \Delta E \leq 0 \end{cases} \tag{5}$$

The melt pool traces the scanning path, and in its wake, the material within the heat-affected zone (HAZ) undergoes spin reassignment according to the mobility calculated in Equation 5.

2.3 Simulation Setup

The MC Potts microstructure simulation requires geometric input describing the melt pool geometry. These geometric parameters can be obtained either through finite element simulations or from experimentally measured values. The top surface of the melt pool can be determined experimentally, but the subsurface geometric values are difficult to infer. Typically, FEM simulations are conducted to estimate the full three-dimensional geometric information. Once the geometric parameters are identified, a Bezier curve is fitted using control points calibrated to the melt pool geometry as described in the previous subsection.

The MC Potts simulation uses rejection KMC with a square lattice constant of $10\mu m$ with non-periodic boundaries to represent a $5mm \times 5mm \times 5.08mm$ section of an additively manufactured component as shown in Figure 2. The constant melt pool has a length, width, and depth dimensions of $3.71mm$, $1.65mm$, and $1.27mm$ respectively. The hatch spacing is $1.92mm$ and the scanning speed is set to $70\frac{\mu m}{MCS}$ aligned parallel to the x-axis. Each simulation uses 1.27×10^8 lattice sites with layers built from the bottom up, mirroring the direct energy deposition process.

3 Results

3.1 Qualitative comparison with EBSD images

Figure 2 illustrates a layer-by-layer build process. Due to the ratio of the hatch spacing to the width in the y-direction, each layer consists of three passes per layer parallel to the x-axis.

The results obtained from the MC Potts simulation are qualitatively compared to experimental results [11]. To compare the microstructure of the printed component, three orthogonal slices are extracted: X-cut, Y-cut, and Z-cut, taken at planes $x = 3mm$ from the $y = 2.8mm$, and $z = 3.3mm$, respectively. Figure 3 presents a qualitative comparison between the simulation and experimental images. The EBSD microstructure maps are shown for all three orientations. The X-cut reveals grains extending from side to side, tracing the melt pool curvature, which is successfully

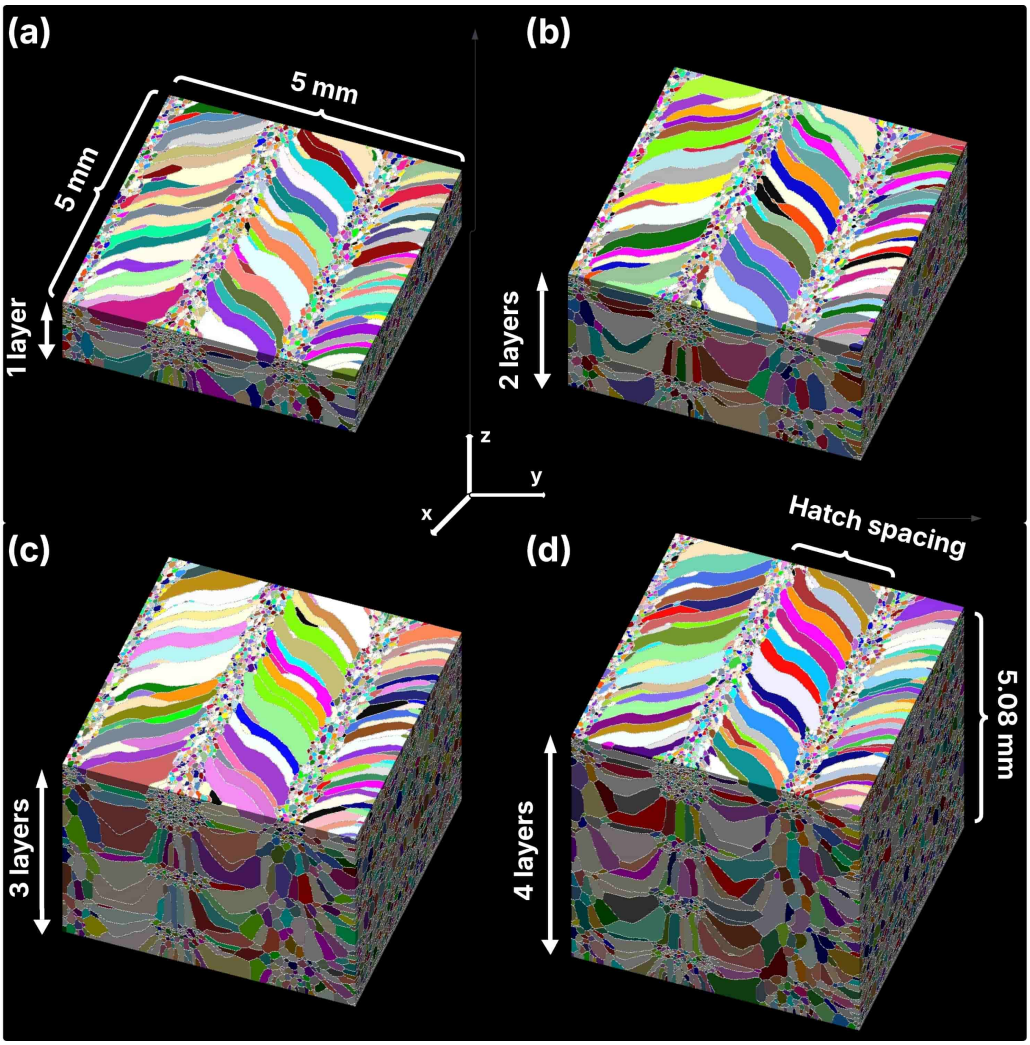


Figure 2: Layer-by-layer material deposition simulated using the MC Potts model. (a) First layer. (b) Second layer. (c) Third layer. (d) Fourth layer.

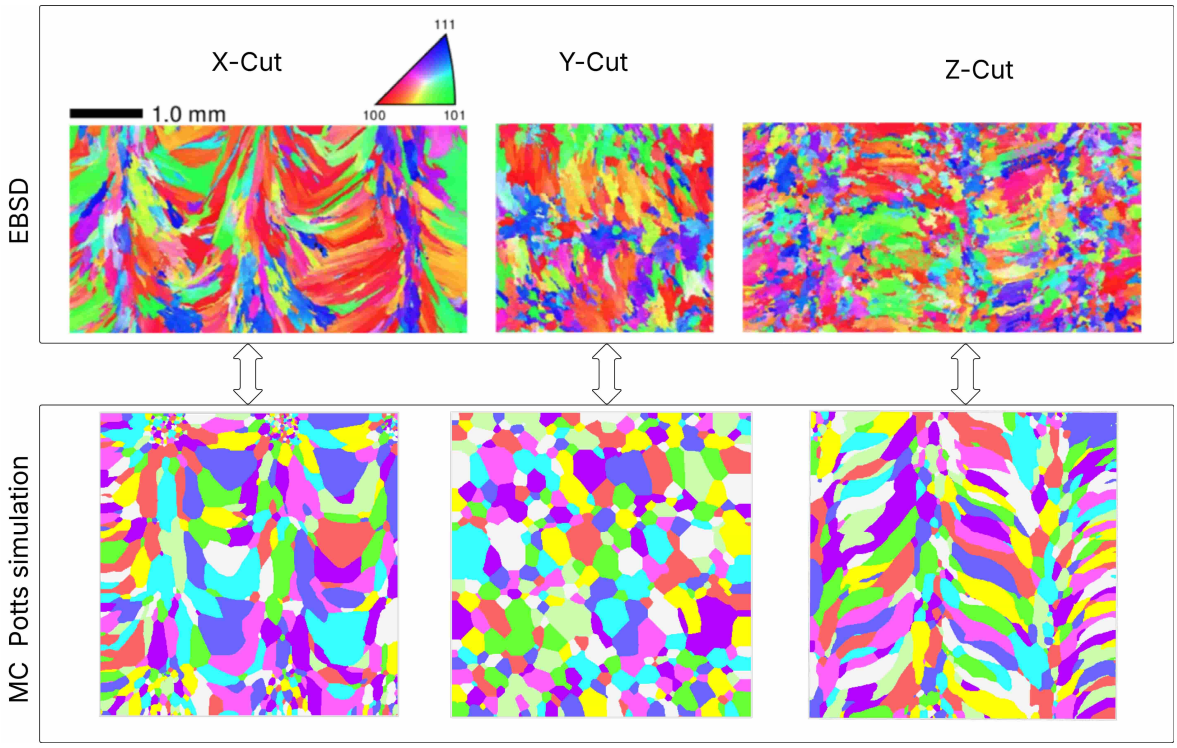


Figure 3: Contrasting the simulated grain structures with EBSD images (adapted from [11]) for the three orthogonal directions, X-cut, Y-cut, and Z-cut.

replicated by the MC simulation. The Y-cut from the EBSD map shows a combination of columnar grains and equiaxed grains. In contrast, the Y-cut from the simulation shows predominantly equiaxed grains rather than the mixed morphology observed in the EBSD map. The Z-cut from the EBSD map shows grains growing in the scanning direction. The simulated Z-cut also demonstrates three alternating scanning directions and how the grain growth follows the scanning path.

3.2 Quantitative comparison with EBSD images

To compare the simulated and the EBSD microstructure maps, the grain area and eccentricity are used as quantitative metrics. Figure 4 provides a quantitative comparison. Comparing the grain areas, the X-cut simulated microstructure shows excellent agreement with the EBSD image. However, performance varies for other orientations. The Y-cut and Z-cut show some deviation, though notably all results remain within the same order of magnitude, demonstrating reasonable predictive capability of the MC Potts Model. Eccentricity is chosen as another criterion for comparing and evaluating grain morphologies. Eccentricity measures how close the grain is to a circle by first fitting the grain to an ellipse and calculating one minus the squared ratio of the minor axis length to the major axis length ($1 - \frac{b^2}{a^2}$). A value near 0 indicates nearly circular grain, while a value close to 1 implies high elongation (a typical property of columnar grains). As shown in Figure 4, the eccentricity for the X-cut matches the EBSD results well.For the Y-cut and Z-cut, the simulation successfully captures the primary trend that a larger area percentage is dominated by grains with higher eccentricity, though with greater variability in the distribution.

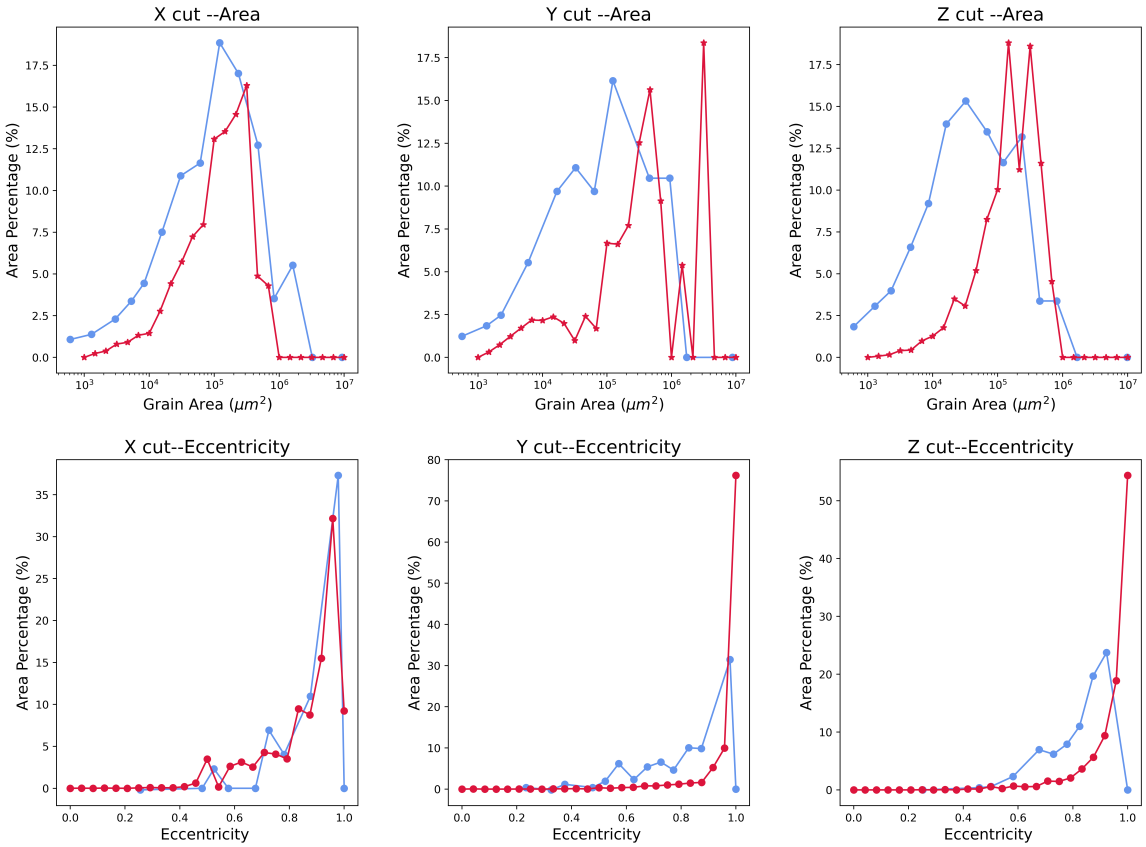


Figure 4: Validation plot comparing the simulated grain structures with EBSD images using grain area (top row) and eccentricity (bottom row).

4 Discussion

Several factors contribute to the observed deviation between the simulated and EBSD microstructures. First, the simulations are stochastic in nature, meaning that individual runs will naturally show some variation from experimental EBSD data unless sufficient simulations are conducted to capture the mean behavior. Additionally, noise may be introduced during EBSD data acquisition and the post-processing methods employed.

The quantitative deviations are more pronounced for the Y-cut and Z-cut as compared to the X-cut (the scanning direction). This occurs because the slicing plane for the X-cut analysis is relatively stable, as it corresponds to a uniform melt pool traveling in the x-direction. In contrast, for the Y-cut and Z-cuts, the slicing position of the cut plays a major role. The grain area and eccentricity distributions vary significantly with position along the y and z directions, making these cuts more sensitive to the exact location chosen for analysis.

4.1 Perturbation from the reference parameters

Once the MC Potts simulations are compared and validated against the EBSD images, it is possible to vary the processing parameters in the MC simulations to explore the model's predictive utility. Here, two parameters are varied and compared against the base case, and their influence on microstructure is assessed. These parameters are the scanning speed and melt pool size. The grain microstructure obtained from these modified simulations is presented in Figure 5.

The first row in Figure 5 shows the results from the MC Potts simulation performed with an increased scanning speed of $120\frac{\mu m}{MCS}$, which is $1.71\times$ that of the base case presented in Figure 3. The grains near the center of the melt pool are relatively smaller and equiaxed; however, significant grain growth occurs farther from the center of the heat source. The second row shows the simulated microstructure obtained with an increased size of the melt pool by a factor of 1.5. The larger melt pool expands the processing zone, enabling extensive grain growth. In the X-cut and Y-cut, there is a significant columnar grain growth spanning over four layers.

4.2 Future improvements

The simulation method implemented in SPPARKS has several limitations stemming from oversimplified assumptions. It assumes the melt pool geometry is constant throughout the simulation since the thermal and grain growth are decoupled. Another important enhancement to the MC Potts model for AM could be crystallographic orientation as a description of the grains, since thermal properties of the material are dependent on grain orientations. The prediction accuracy of simulated grain growth, when compared to experimental observations, can be enhanced by utilizing machine learning techniques. Deep learning algorithms have previously shown promising results in predicting grain growth in metals [12].

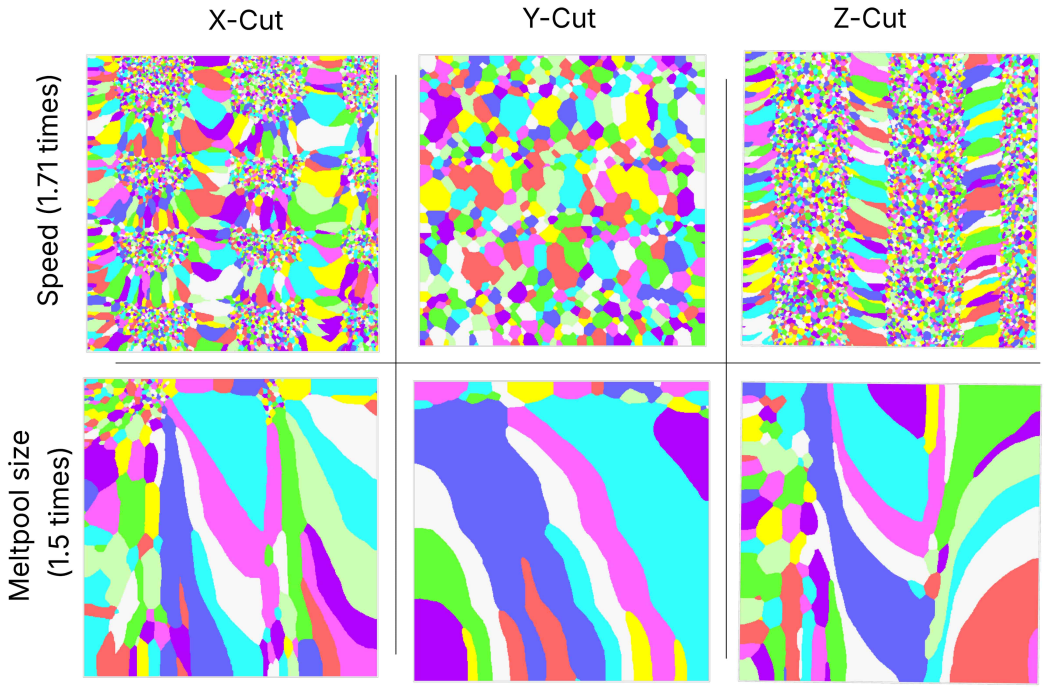


Figure 5: Results of the perturbation of the processing parameters. The first row shows grain structures with an increased scanning speed. The second row shows grain structures with an expanded melt pool.

5 Conclusion

This work demonstrates the application of MC Potts simulation using SPPARKS software to model microstructure evolution in 304L stainless steel additively manufactured by the Direct Energy Deposition method. The simulation results successfully replicate key microstructural features observed by EBSD maps, with the X-cut showing excellent agreement in both grain area and eccentricity. While the Y-cut and Z-cut exhibit greater variability, the simulations capture the dominant trends and maintain results within the same order of magnitude as experimental observations. Parametric studies reveal that increasing scanning speed produces smaller, more equiaxed grains near the melt pool center, while increasing melt pool size promotes extensive columnar grain growth spanning multiple layers. Despite limitations from simplified assumptions such as constant melt pool geometry and decoupled thermal-microstructure models, the MC Potts approach provides a computationally efficient framework for predicting microstructure evolution in metal additive manufacturing.

Data availability

The data and Python scripts used for analysis are available at GitHub: `https://github.com/Wongel-Yilma/AM_microstructure_simulations`.

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